

ECSE-4040

Digital Electronics

Lecture 2

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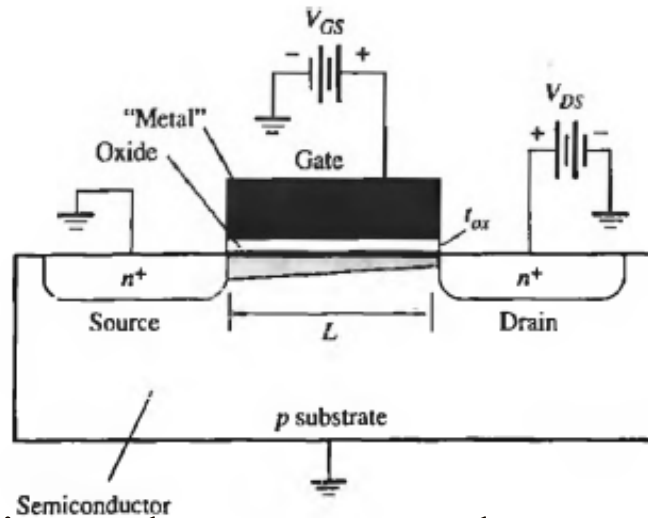
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<https://lms.rpi.edu>

Lecture Outline

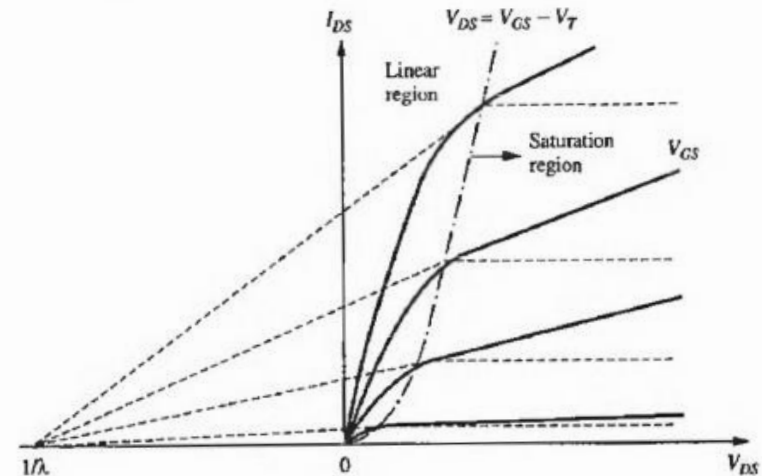
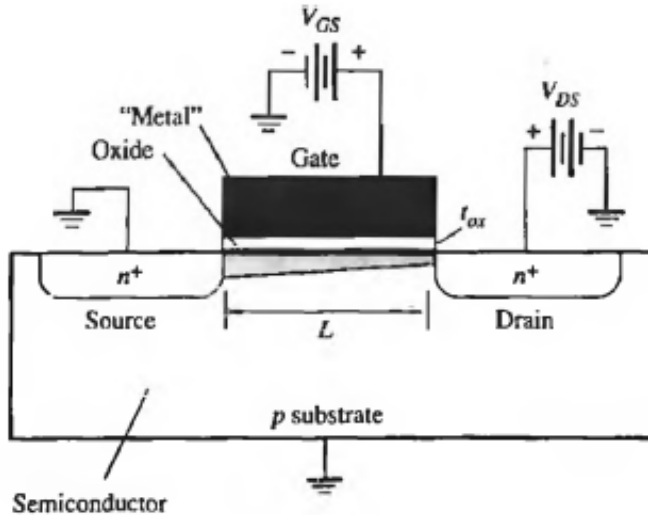
- MOSFET
 - NMOS vs. CMOS
 - Interconnect vs. Gate Delay
 - Technology and Structural Evolution
- Digital MOS IC Design Trends

MOSFET



- MOSFET – Transistor that uses a voltage, through a capacitor, at the controlled terminal (gate) to control the current flow between two main terminals
- Threshold Voltage (V_T): Boundary between ON and OFF states
- Channel Type: N Channel (e^-) vs. P Channel (h^+); N-Channel preferred

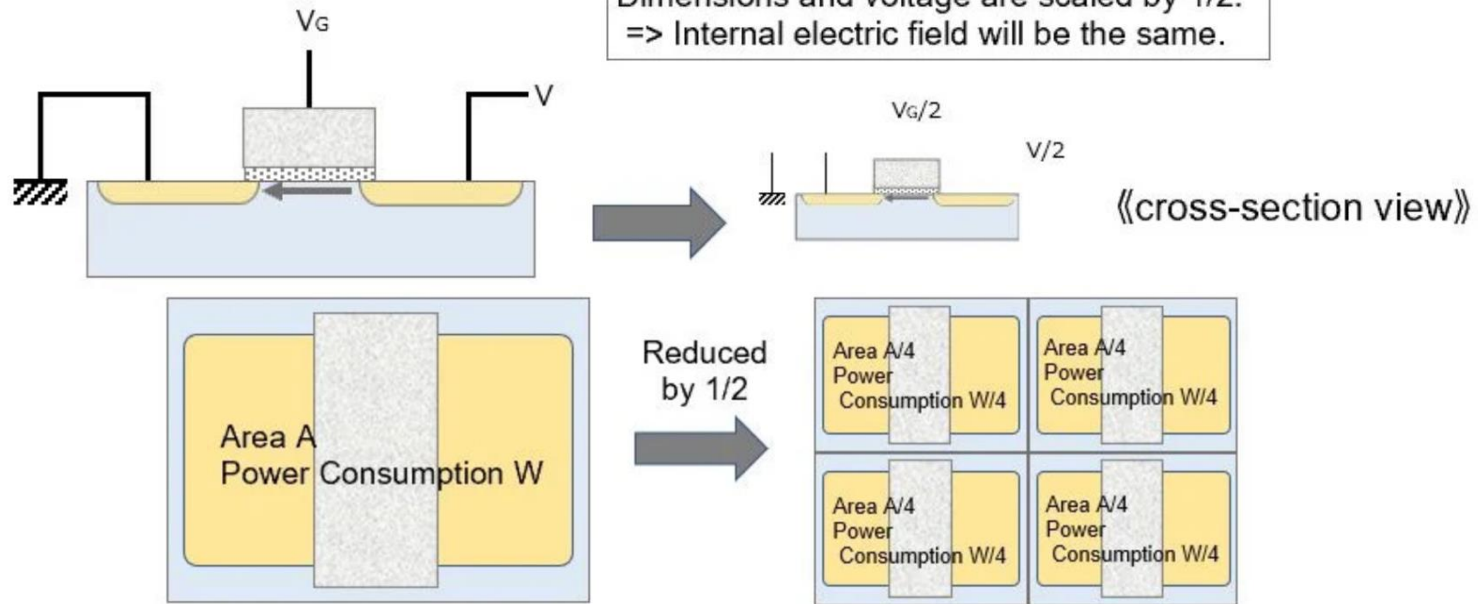
MOSFET I-V Characteristics



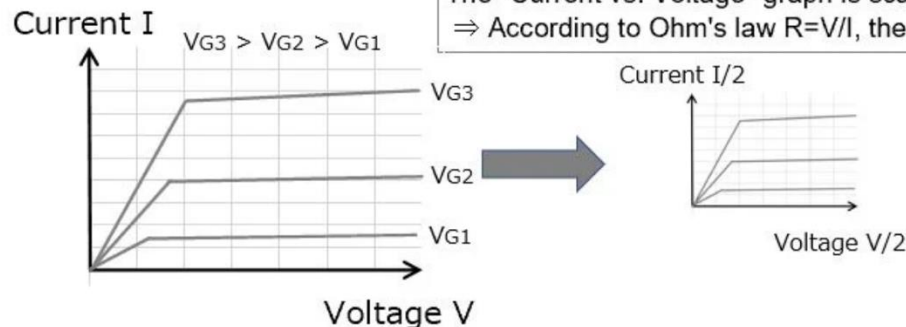
- Transistor behaves either as a gate-controlled resistor (linear) or current source (saturation)
- Inversion channel carriers reach maximum, sometimes saturation, velocity at the drain end
- $I_{DS} = f((W/L), \mu \text{ or } v_{sat}, C_{ox}, V_{GS}, V_{DS}, V_T)$
 $I_{DS} \propto (V_{GS} - V_T), V_{DS}$ linear
 $I_{DS} \propto (V_{GS} - V_T)^2$ saturation

MOSFET Scaling

Dimensions and voltage are scaled by $1/2$.
 $\Rightarrow$ Internal electric field will be the same.



Theoretical calculations show that the current is also scaled by $1/2$.
 The "Current vs. Voltage" graph is scaled by $1/2$.
 $\Rightarrow$ According to Ohm's law $R=V/I$, the resistance will be the same.



CMOS vs. NMOS Inverters

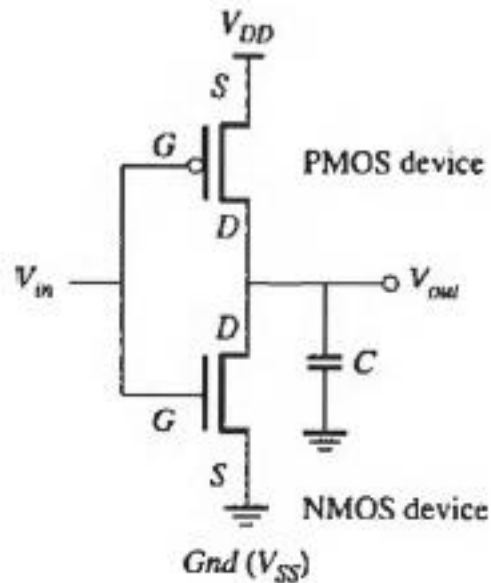


Figure 1.11

CMOS inverter schematic.

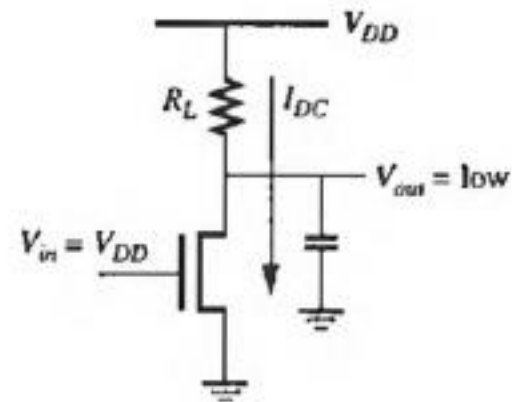
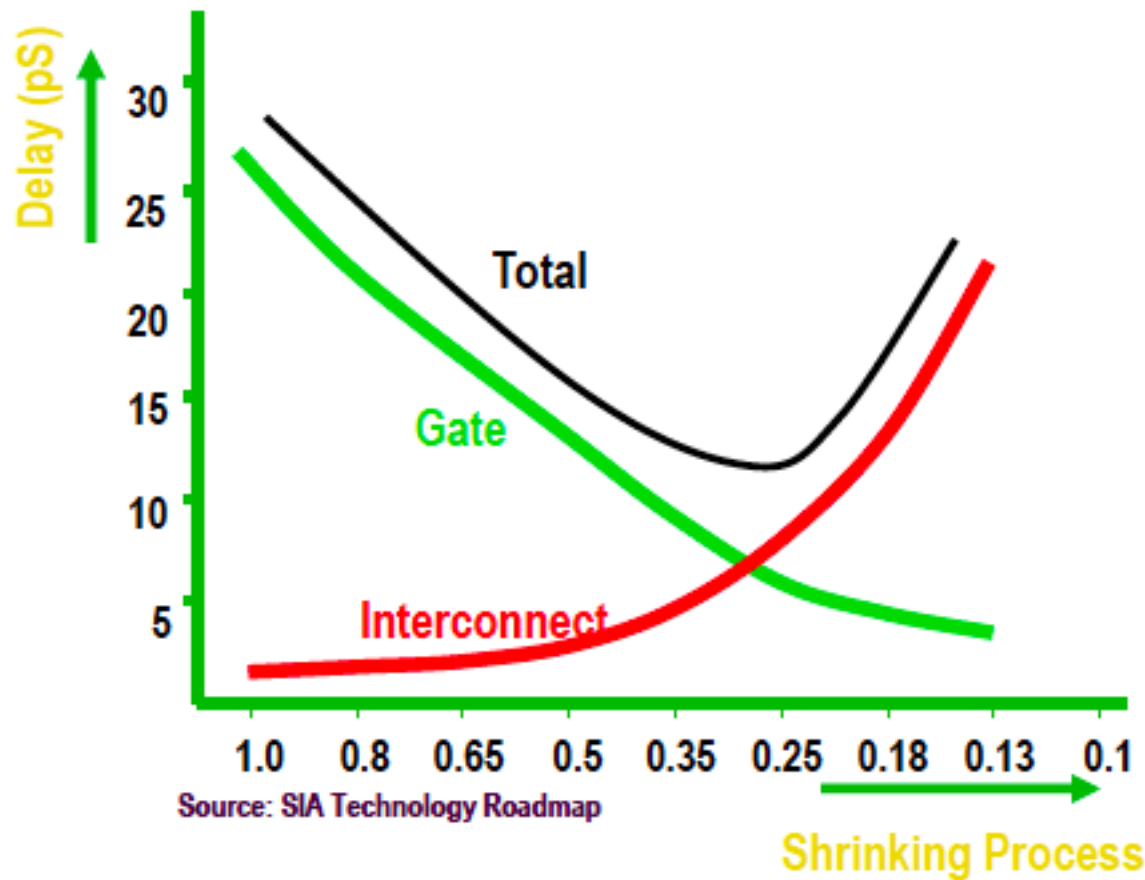


Figure 1.12

Basic NMOS-style inverter.

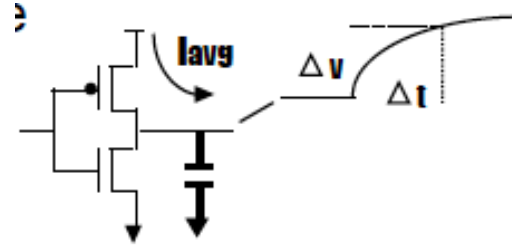
- CMOS: NMOS-Pulldown, PMOS Pullup
- CMOS consumes less power than NMOS and hence becomes dominant

Interconnect vs. Gate Delay

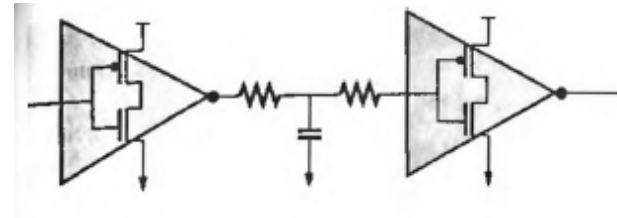
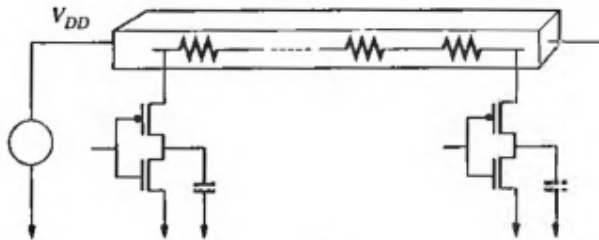


Technology Scaling Effects

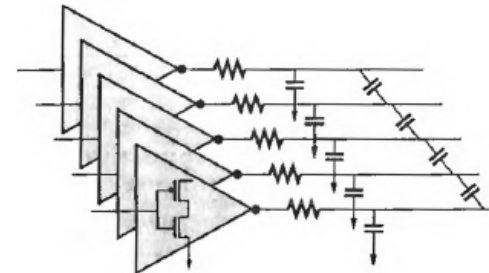
- $\geq 0.5\mu\text{m}$: Gate Capacitance



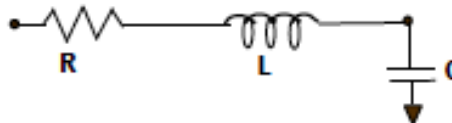
- $\leq 0.35\mu\text{m}$: Resistance



- $\leq 0.18\mu\text{m}$: Coupling Capacitance

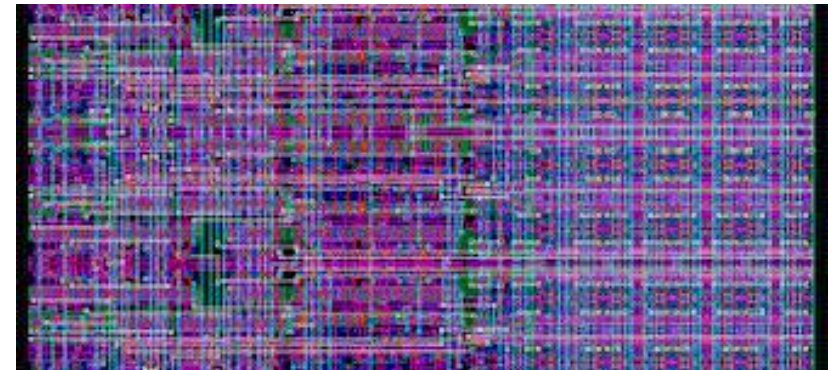
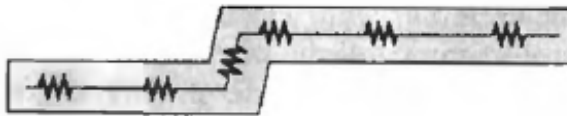
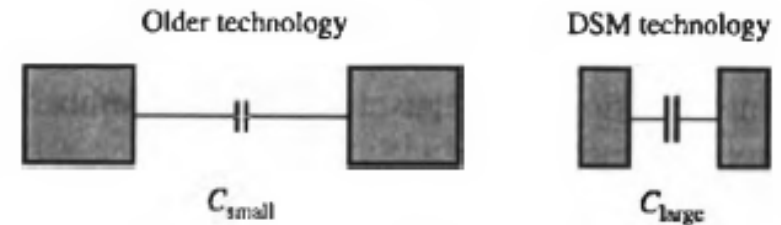


- $\leq 0.10\mu\text{m}$: Inductance



Deep Submicron Interconnect

- Dense Multilayer Metallization increases coupling capacitance
- Long and narrow metal line widths further increase interconnect resistances



Interconnect Delay

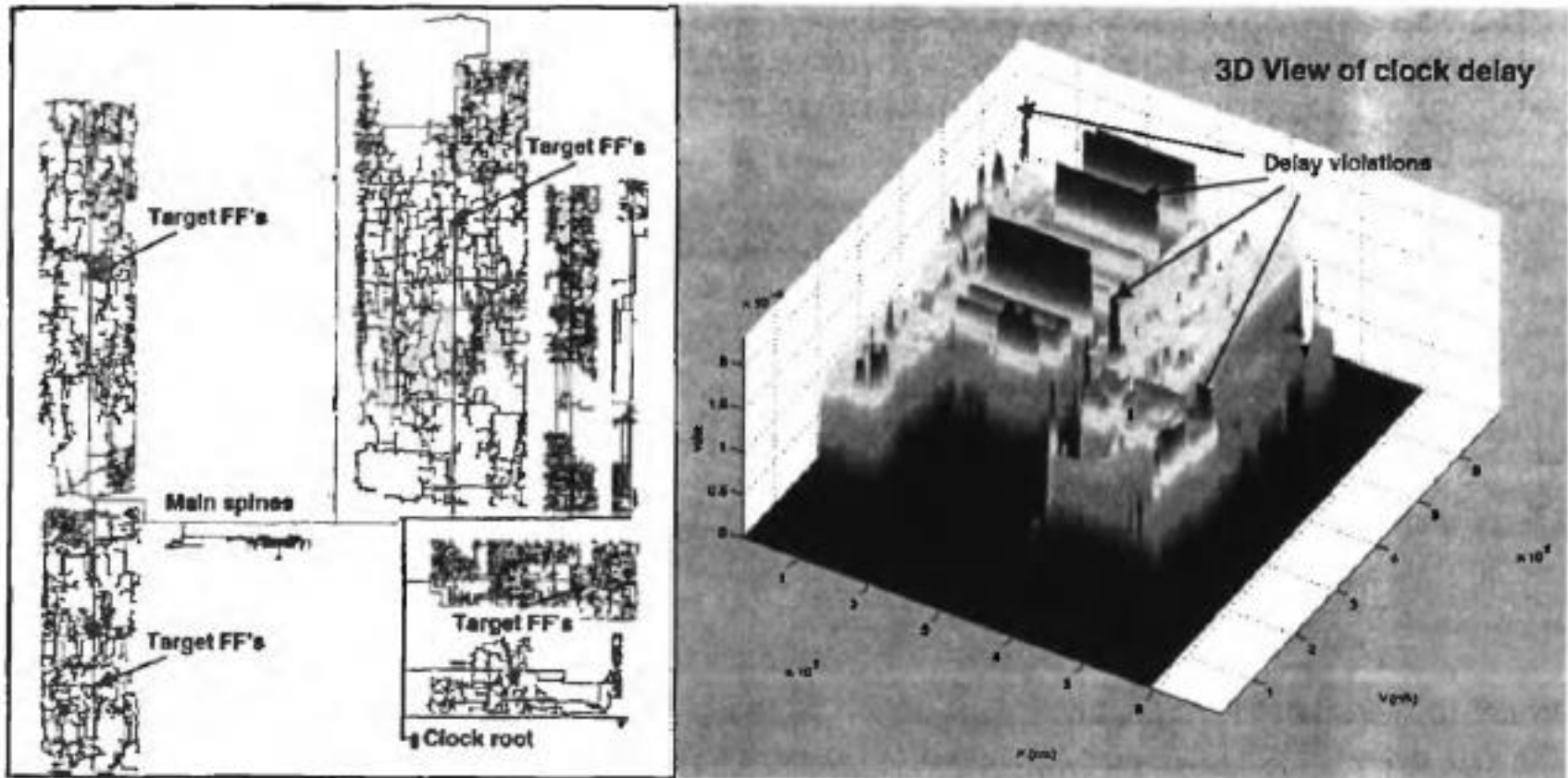


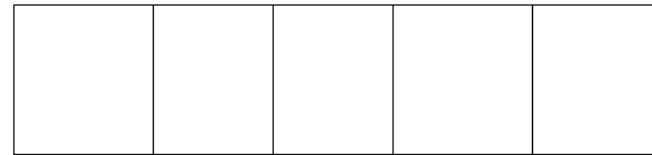
Figure 1.16

Clock tree and 3D delay plot. Plots courtesy of Simplex Solutions, Inc.

IR Drop Problem: Global Routing

- Blocks arrangement and global routing determines IR drops
- IR drop impacts gate timing since it reduces device drive capability
- $3.3V \rightarrow 0.7ns$
 $2.9V \rightarrow 0.5ns$

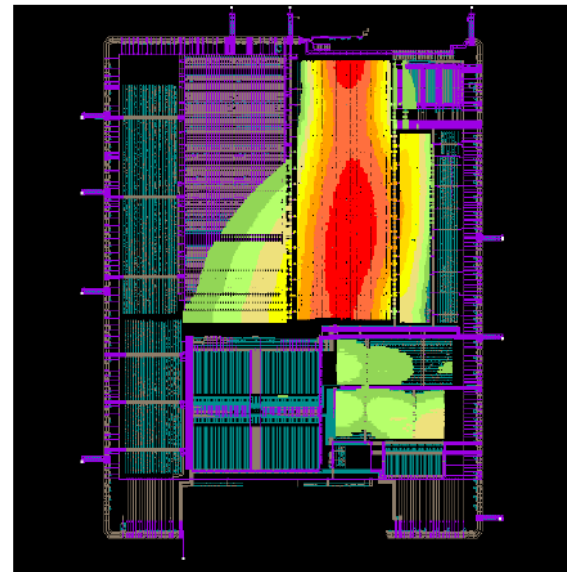
Metal Line for Power Bus



5 sq.



35 sq.



DSM Physical Design Flow

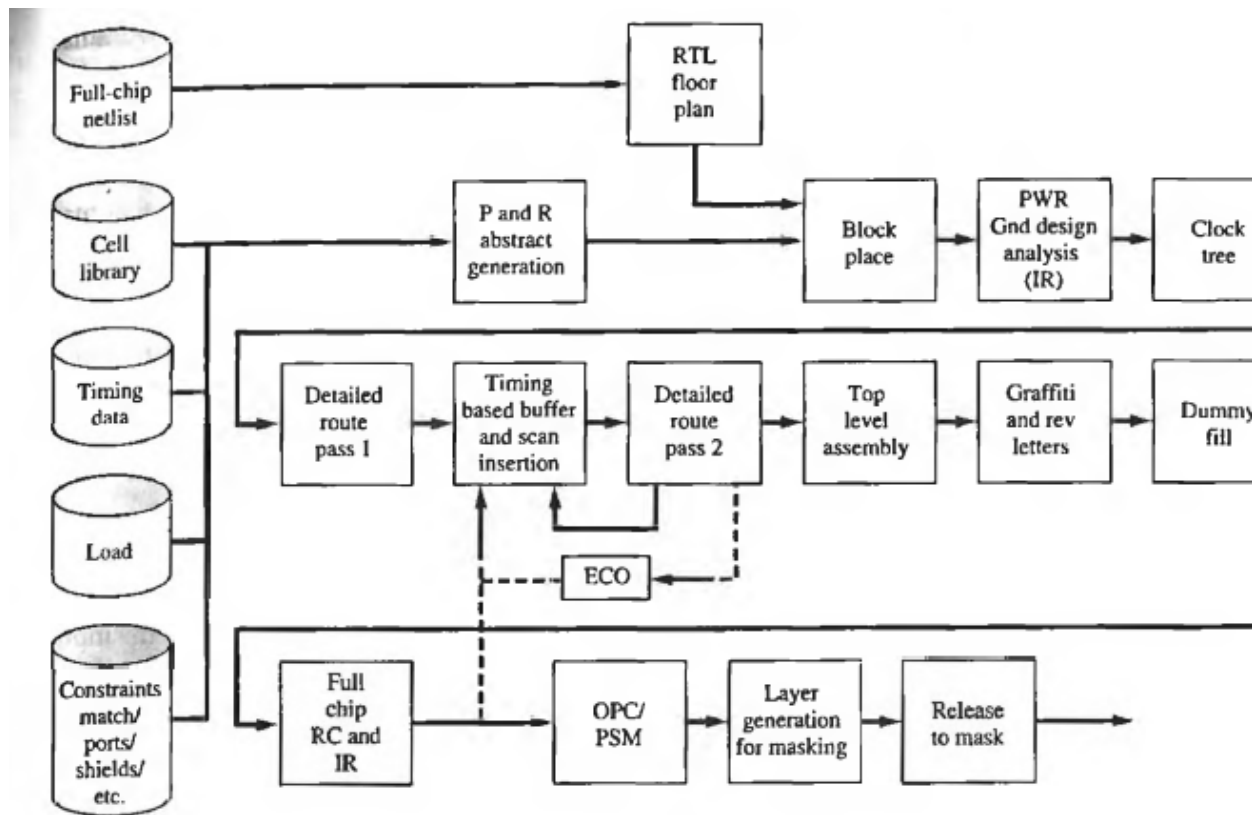
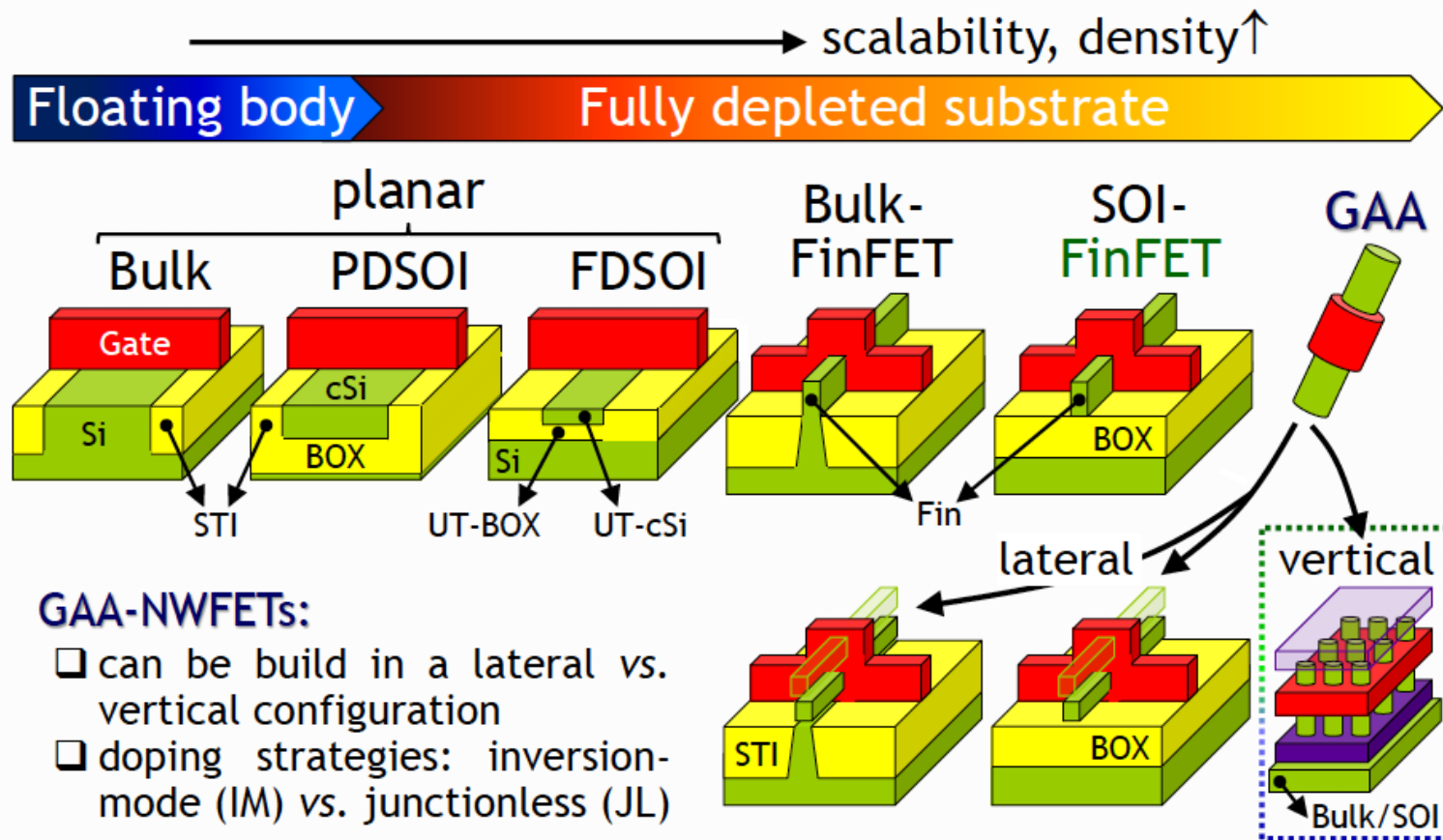


Figure 1.20

Deep submicron physical design flow. Flow courtesy of Silicon Map, LLC.

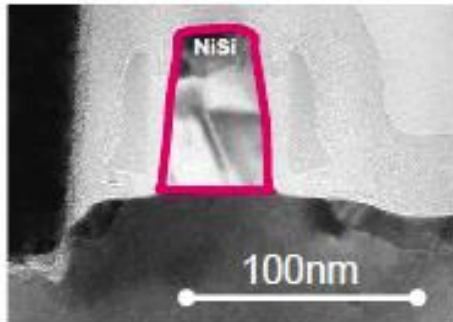
MOSFET Evolution

Introduction: *scaling scenario for device architectures*

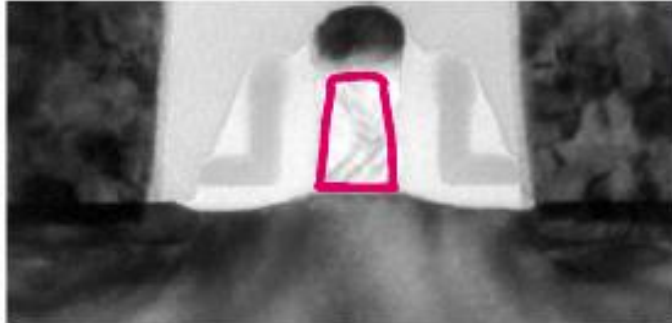


MOSFET Evolution

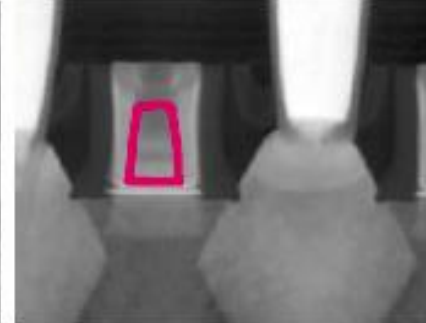
90nm node



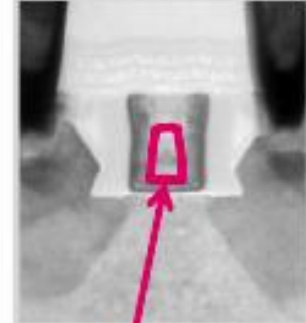
65nm node



45nm node

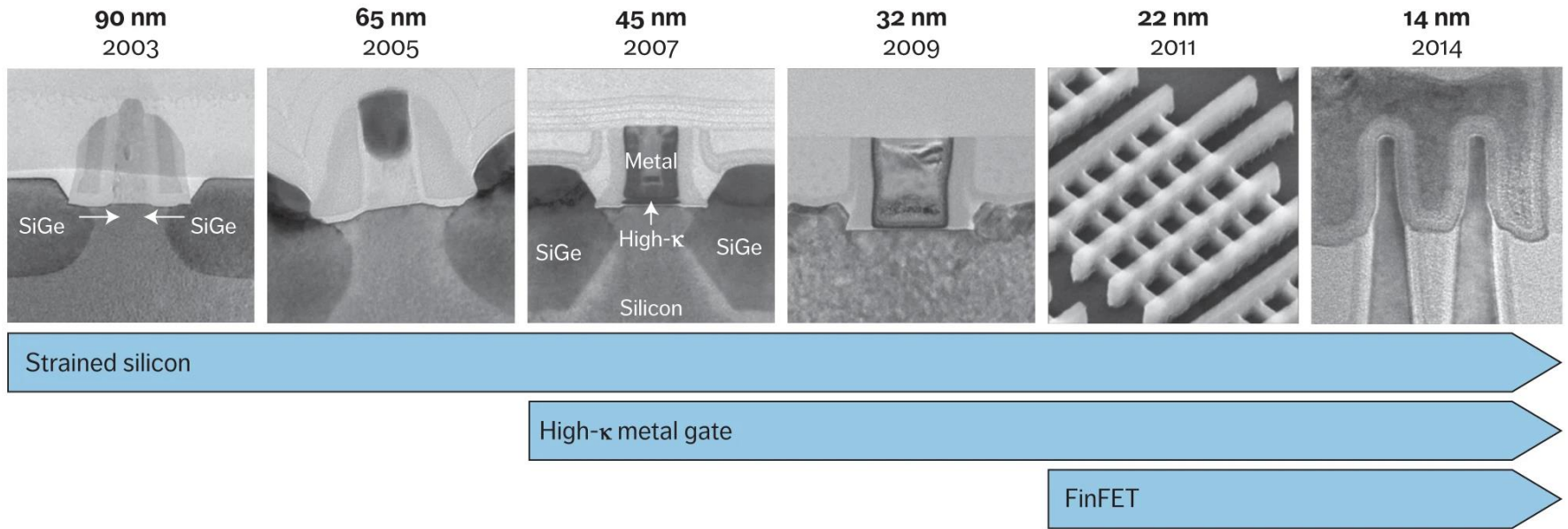


32nm node



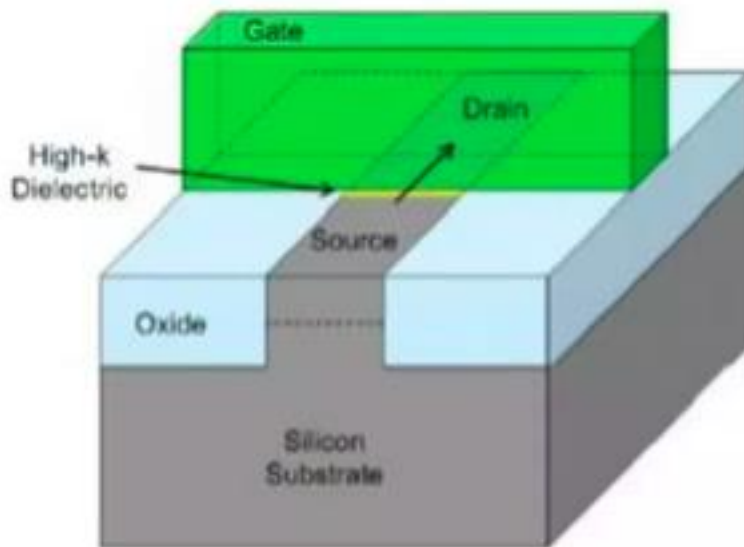
0.7x scaling

MOSFET Evolution



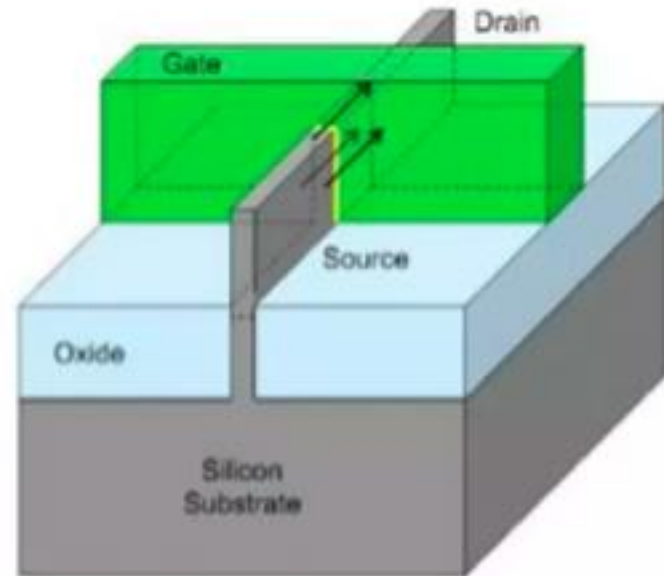
Lateral Planar vs. 3D FinFET

Traditional Planar



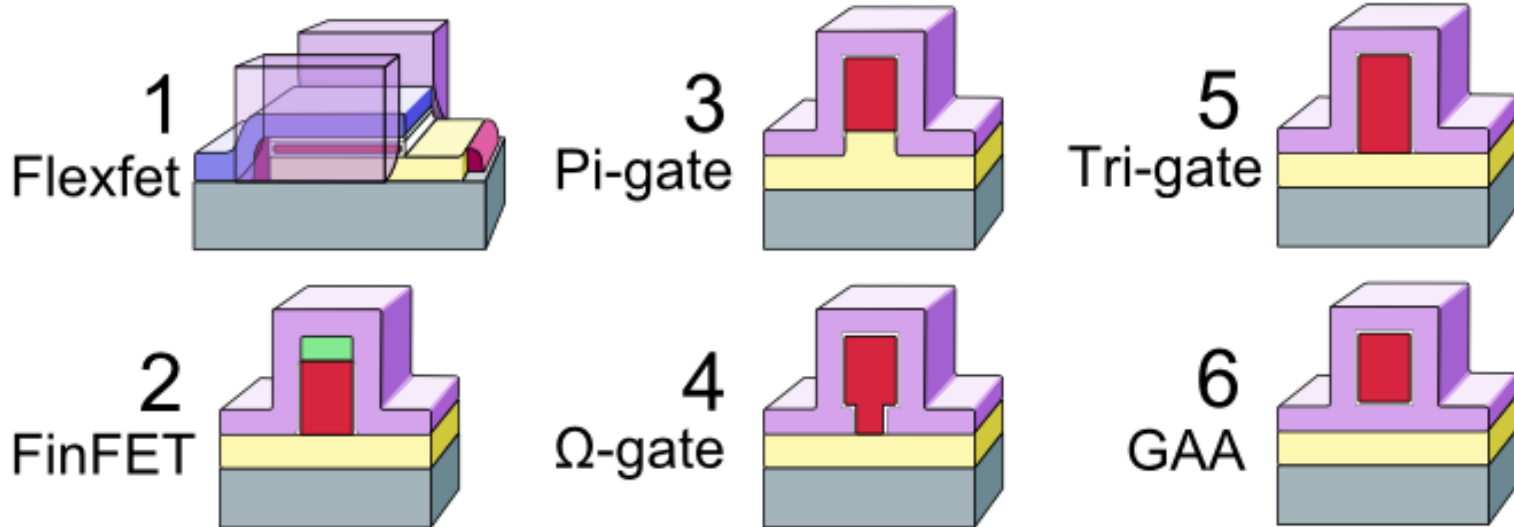
Traditional 2-D planar transistor form a conducting channel in the silicon region under the gate electrode when in the "on" state

3D FinFET



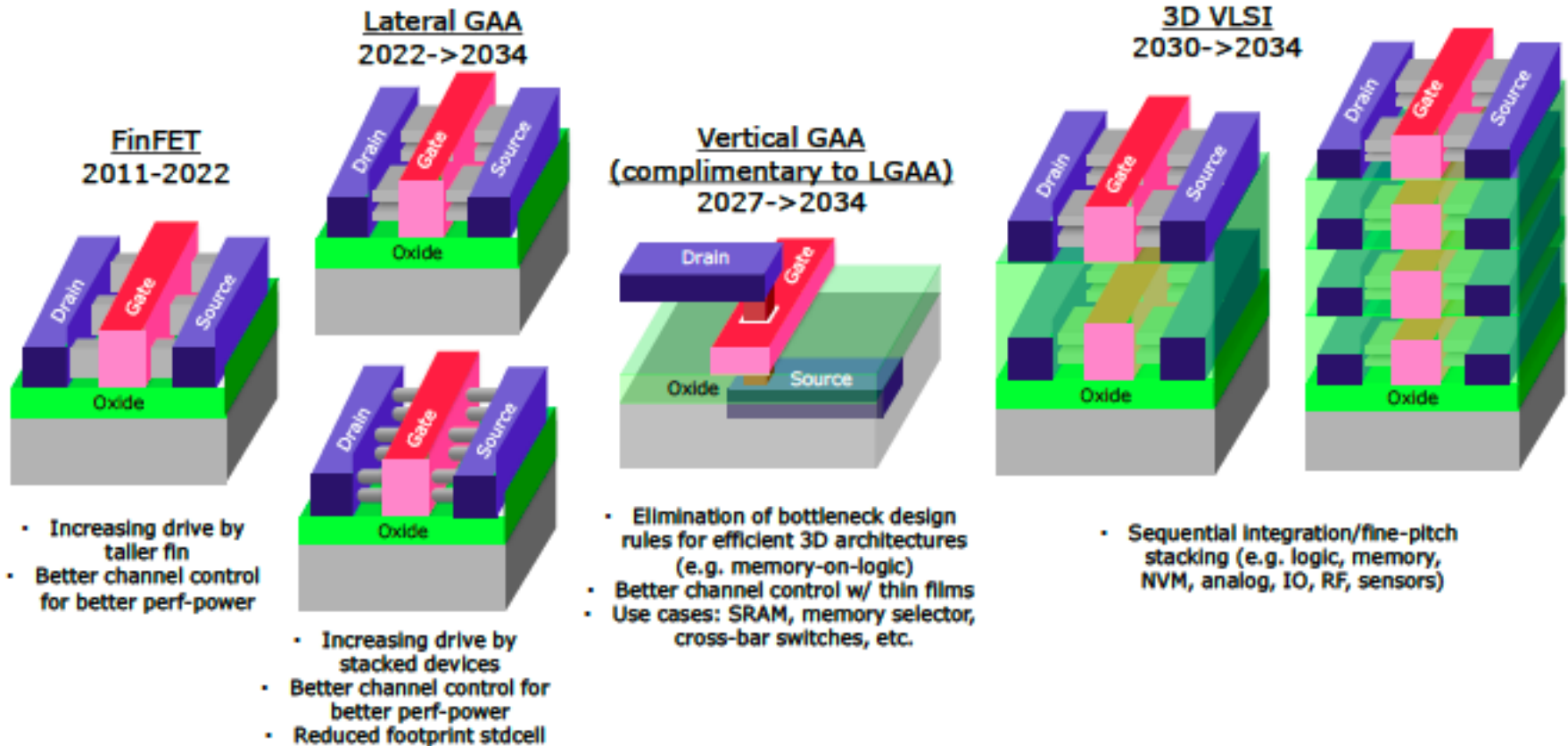
3-D Tri-Gate transistor form conducting channels on three sides of a vertical fin structure, providing "fully depleted" operation

Multi-Gate MOSFETs



MOSFET Evolution

>2020: 2.5D/3D fine-pitch assembly + stacking



Design Trends

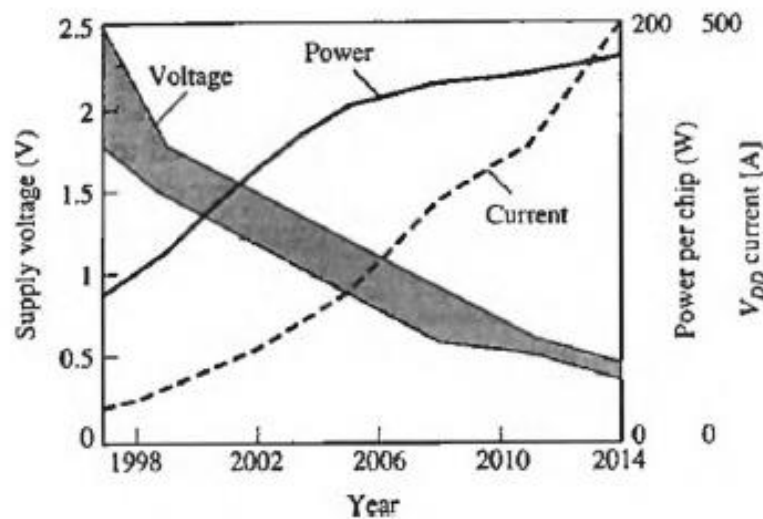


Figure 1.21
Supply voltage and current, chip power trends from ITRS.

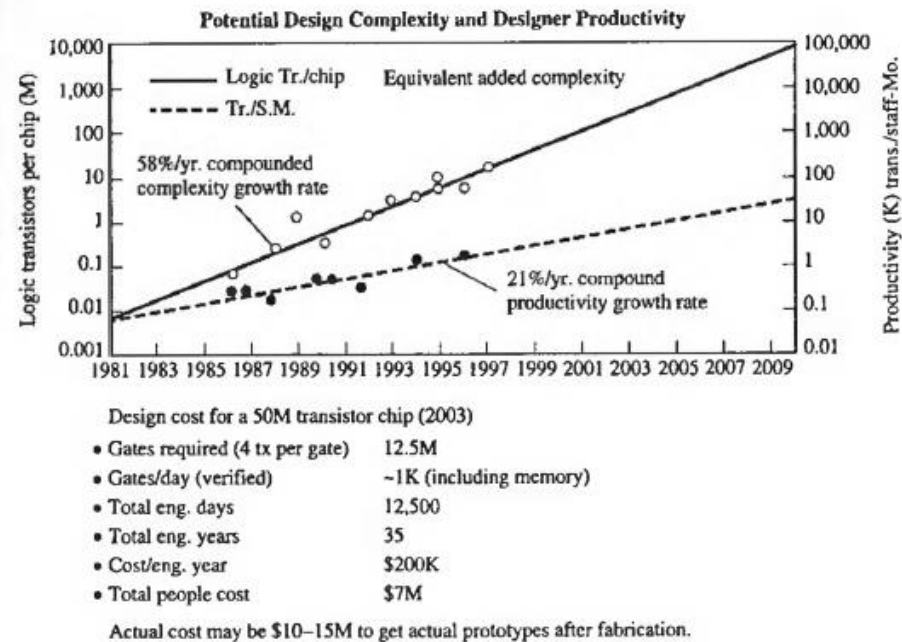


Figure 1.22
ITRS productivity chart and design cost.

System-On-Chip Design

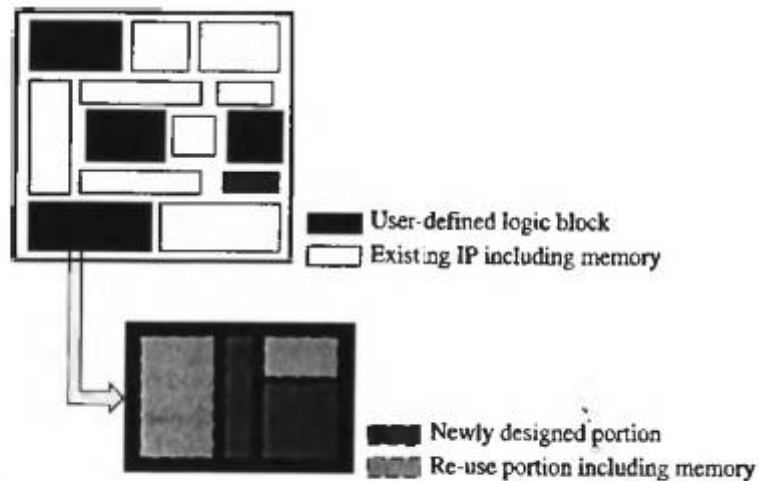
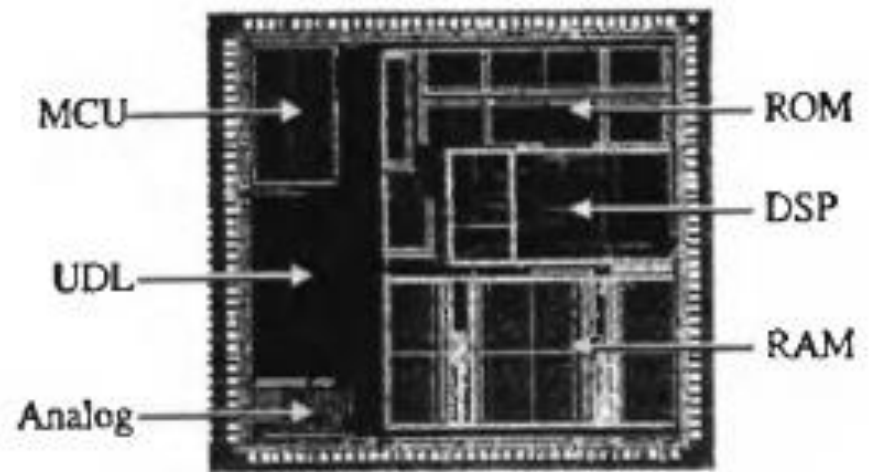
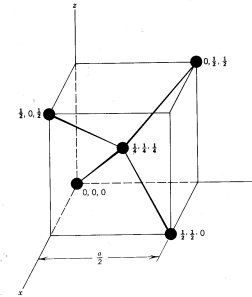


Figure 1.23
System-on-chip design hierarchy



Carrier Concentrations in Semiconductors

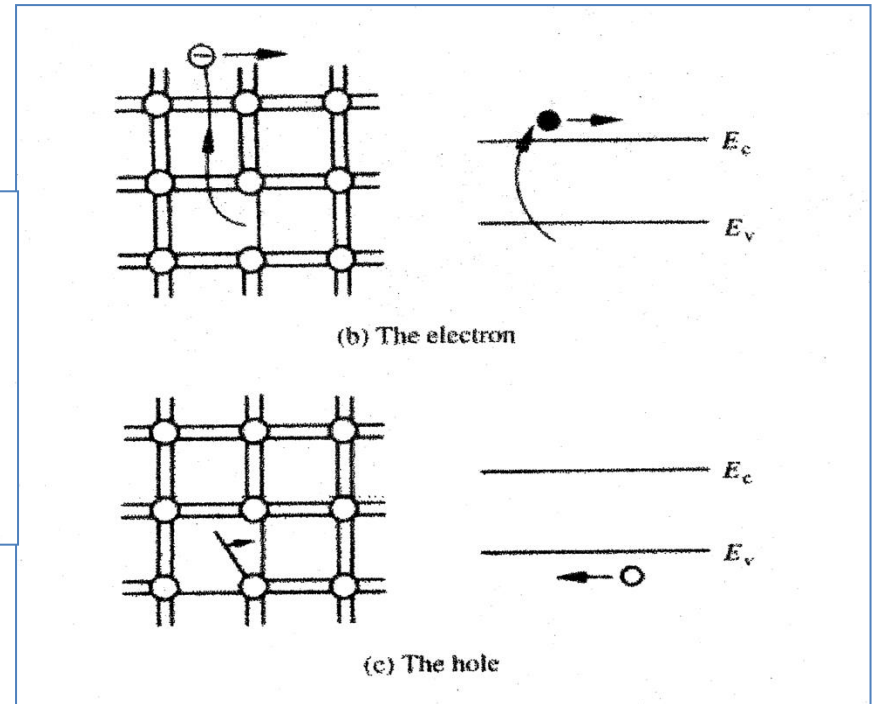
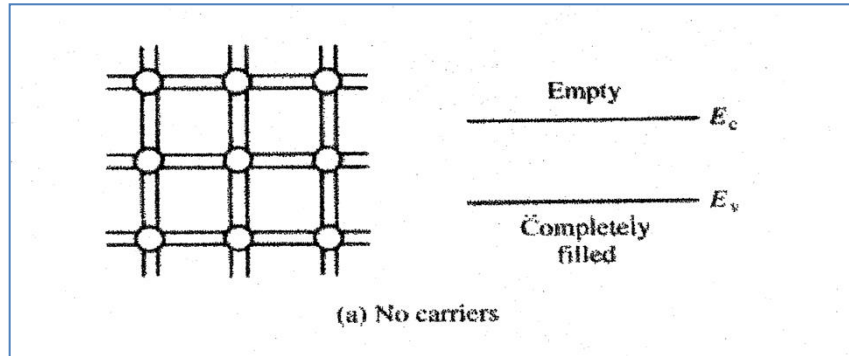


- Crystalline Si has a tetragonally bonded “diamond” lattice.
- n-type Si is doped with donor-atom impurities, such as phosphorus (P). These impurities are $\sim 10^{16} - 10^{20} \text{ cm}^{-3}$. Atomic concentration of pure crystalline Si is $\sim 10^{22} \text{ cm}^{-3}$.
- p-type Si is doped with acceptor-atom impurities, such as boron (B).
- We have: $N_D \rightarrow N_D^+ + e^-$ and $N_A + e^- \rightarrow N_A^- + p^+$. In most cases (extrinsic temperature regime), all initially neutral donor and acceptor atoms are ionized. Thus,

$$n \approx N_D, \quad \text{n-type material}; \quad p \approx N_A, \quad \text{p-type material} \quad (2.1)$$

- N_D is the neutral donor-dopant atom density, N_A is the neutral acceptor-dopant atom density, n is the (free-) electron concentration, p is the (free-) hole concentration. $[n, p, N_A, N_D] = \text{cm}^{-3}$.

Electrons and Holes



Mass Action Law

- Equilibrium statistical mechanics (nondegenerate, Boltzmann limit) allows one to relate electron and hole concentration in a doped (or undoped, intrinsic) semiconductor. The following relations only hold in thermal equilibrium:

$$np = n_i^2, \quad (2.2)$$

where n_i is the intrinsic carrier concentration. At room temperature, $T = 300\text{k}$, $n_i = 1.45 \times 10^{10} \text{cm}^{-3}$.

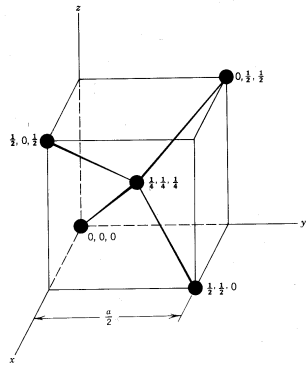
- Therefore, n and p can be computed, given either acceptor or donor dopant concentrations N_A or N_D , respectively.

Carrier Concentrations in Semiconductors

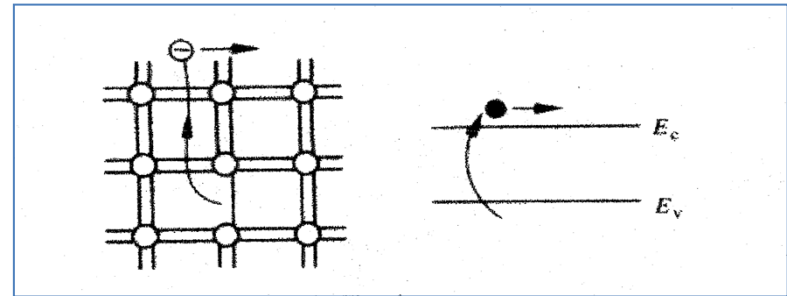
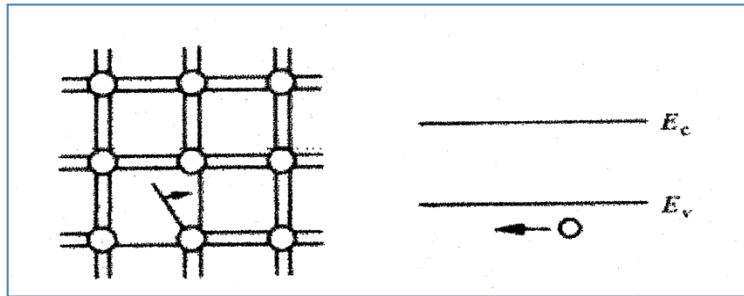
Dopant	Electron Density	Hole Density
Donor ($\bar{N}_D$, n-type)	$n \approx \bar{N}_D$	$p \approx n_i^2 / \bar{N}_D$
Acceptor (N_A , p-type)	$n \approx n_i^2 / N_A$	$p \approx N_A$

Table 2.1: Equilibrium formulas for electron and hole concentrations in a nondegenerate, extrinsic semiconductor.

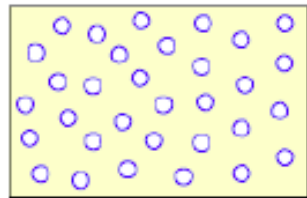
- If $n \gg p$, the material is *n-type*; $p \gg n \Rightarrow$ *p-type*.
- In n-type material, electrons are *majority* carriers and holes are *minority* carriers. In p-type material, *vice versa*.



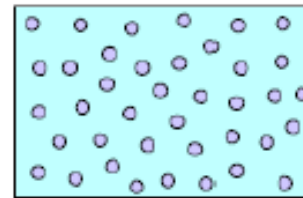
PN Junction and Diodes



p-type semi-conductor heavily doped with acceptor atoms, e.g. boron



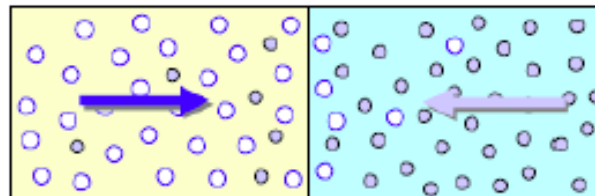
p-type



n-type

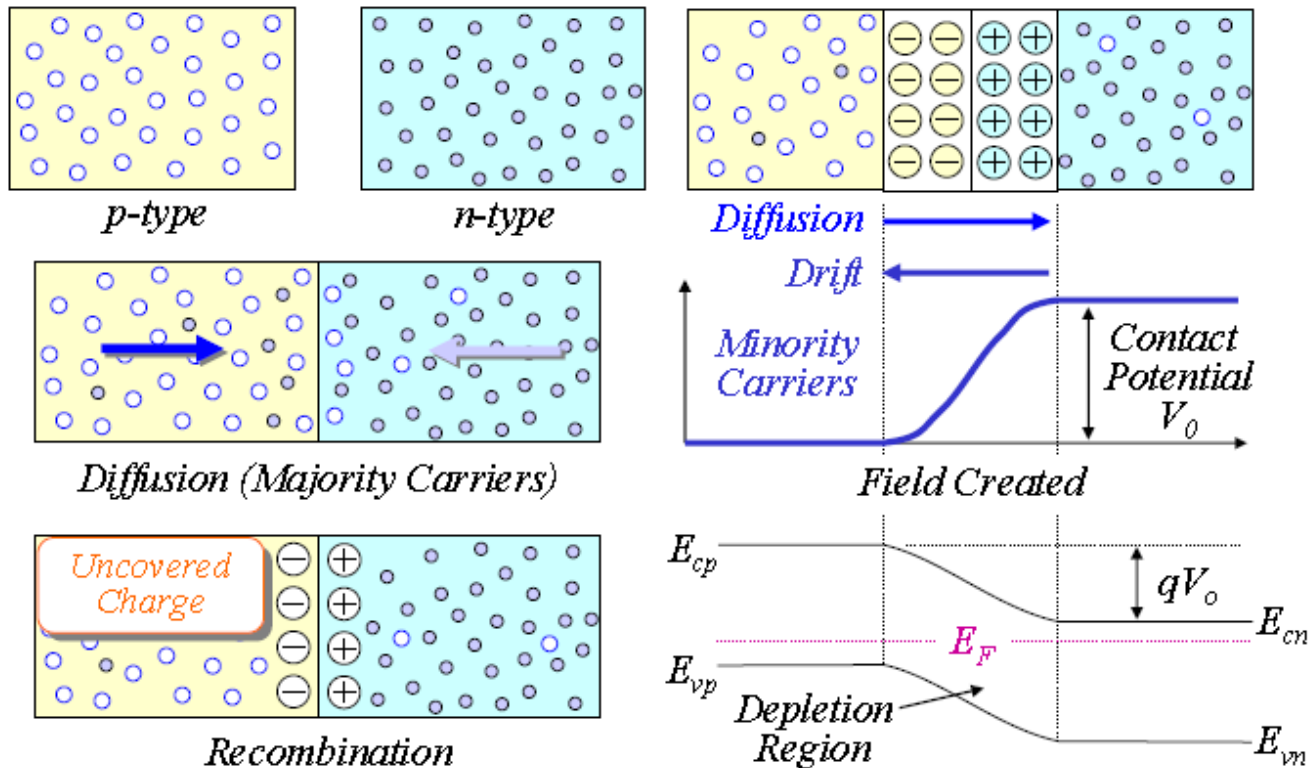
n-type semi-conductor is heavily doped with donor atoms, e.g. arsenic and phosphorus

PN Junction



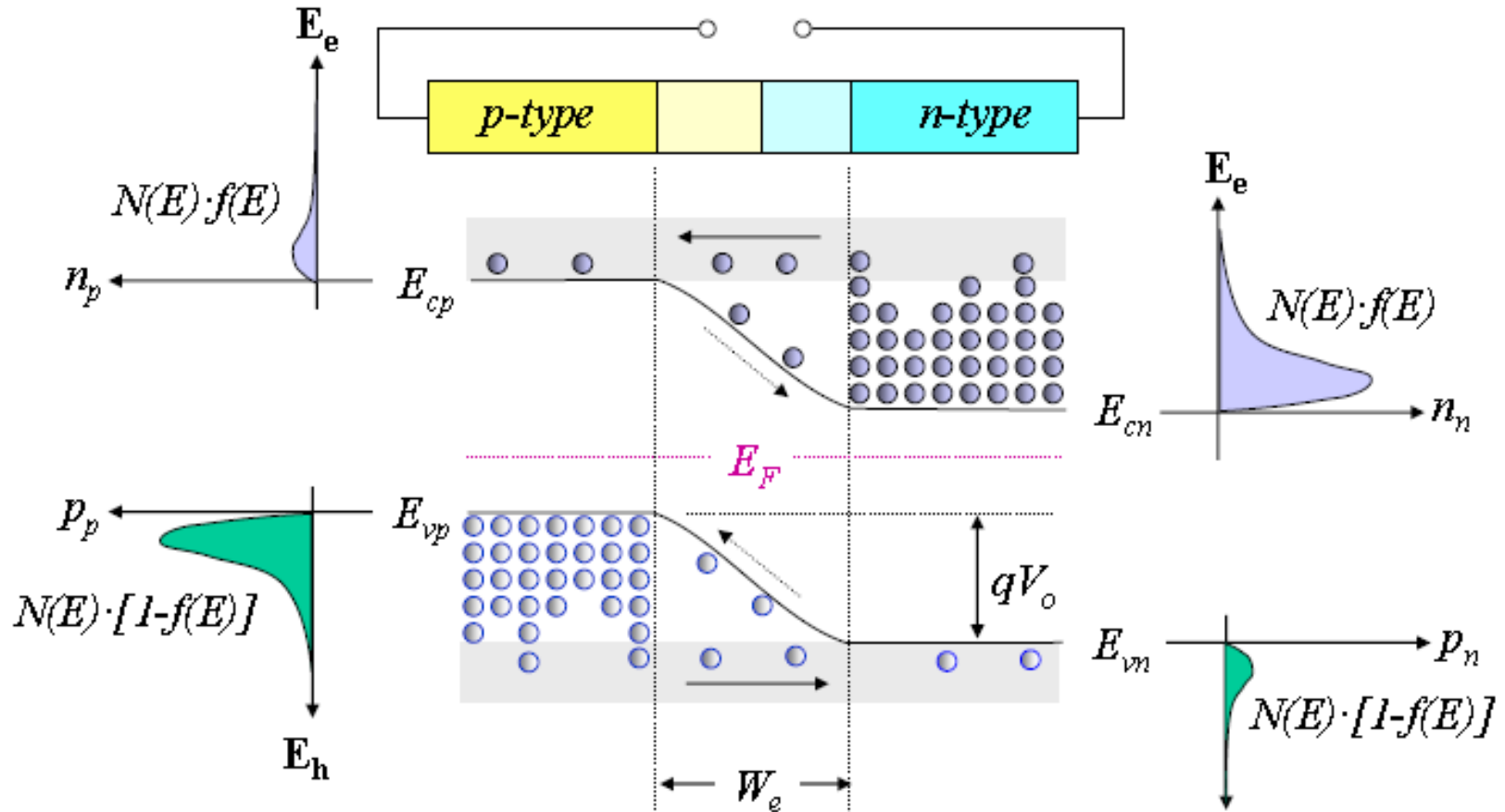
Diffusion (Majority Carriers)

PN Junction



- Depletion Region and Built-in Potential
- Drift and Diffusion Currents
- Asymmetric current flow when biased

PN Junction – Thermal Eq.



- Majority and Minority Carriers

NMOSFET Structure and Operation

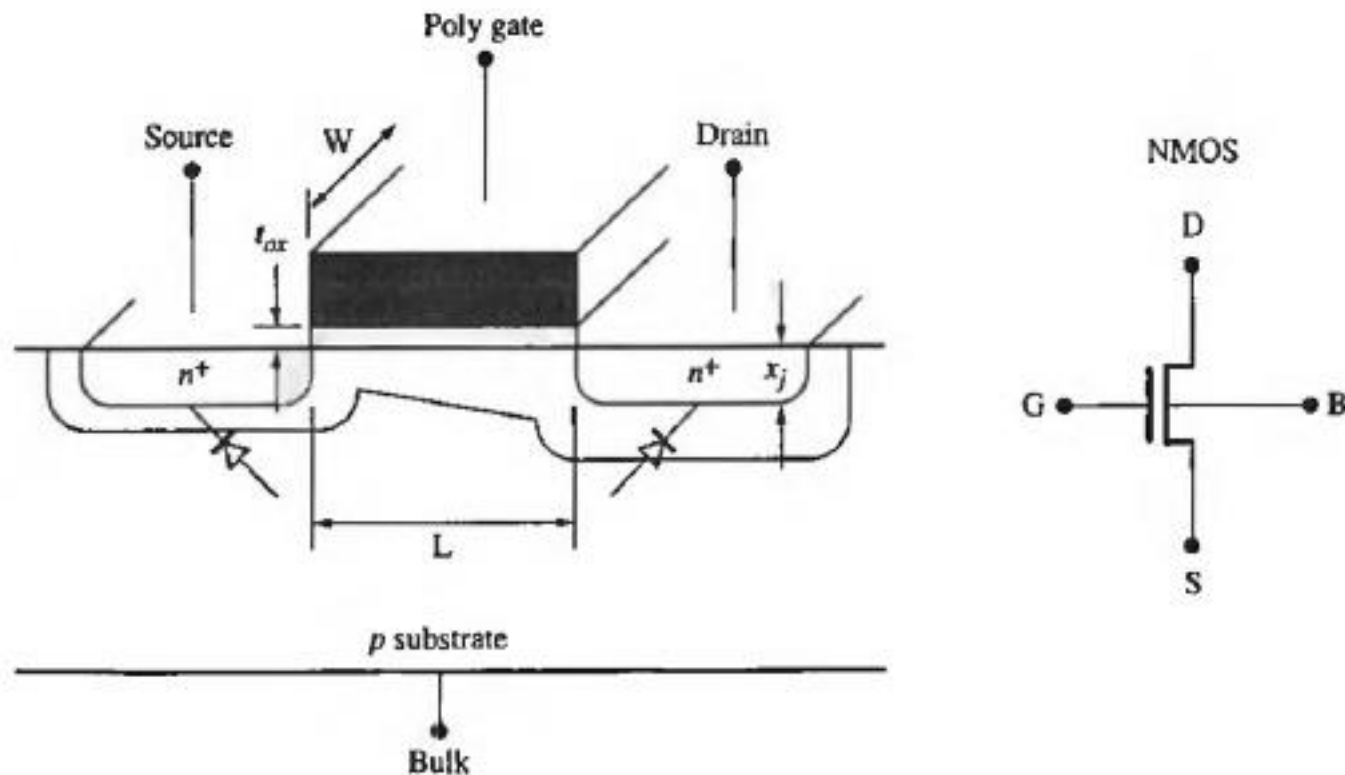


Figure 2.1
NMOS transistor structure and symbol.

PMOSFET Structure and Operation

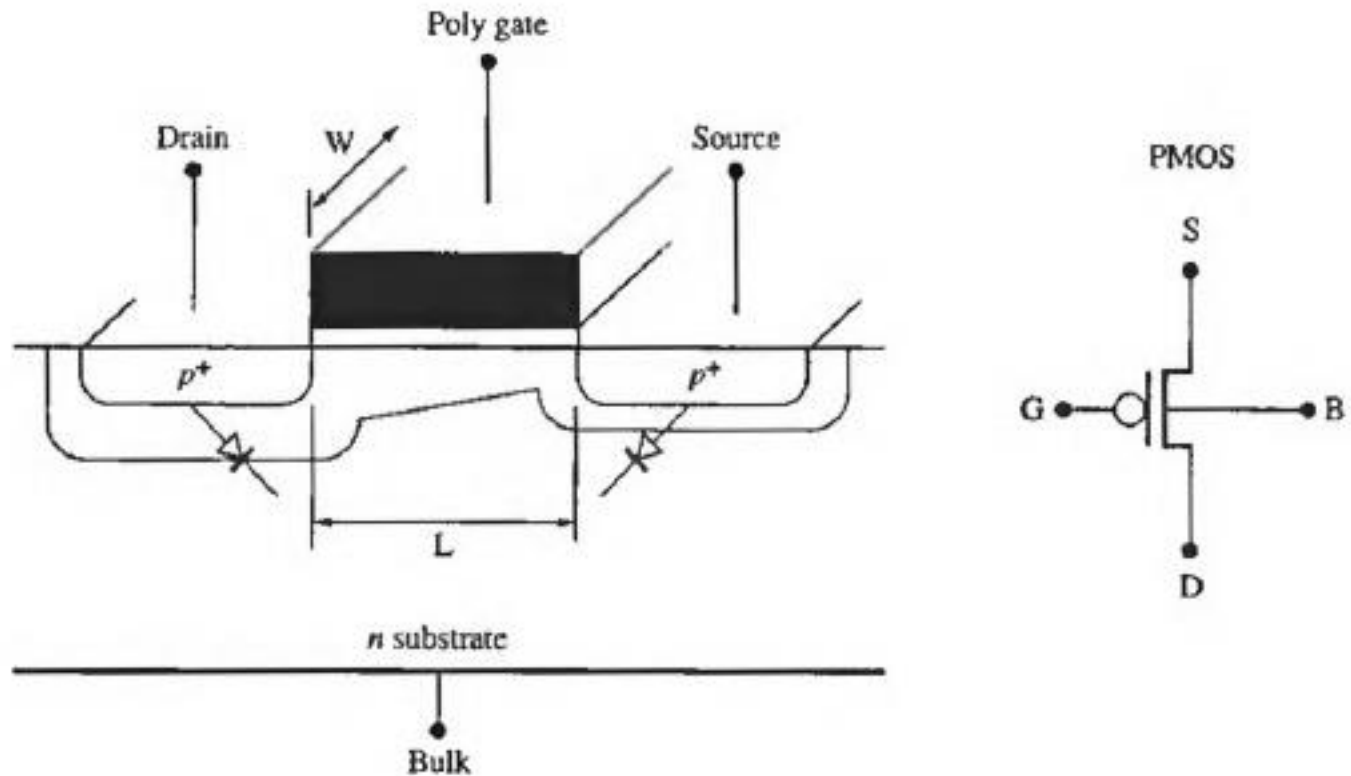
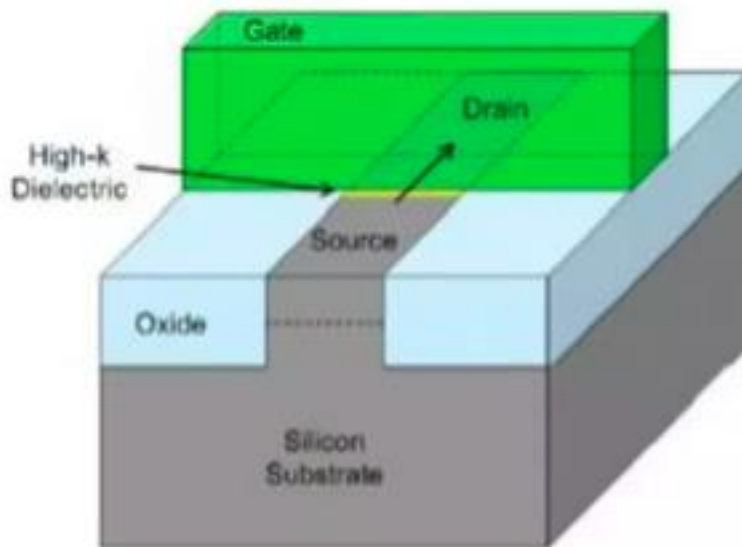


Figure 2.2
PMOS transistor structure and symbol.

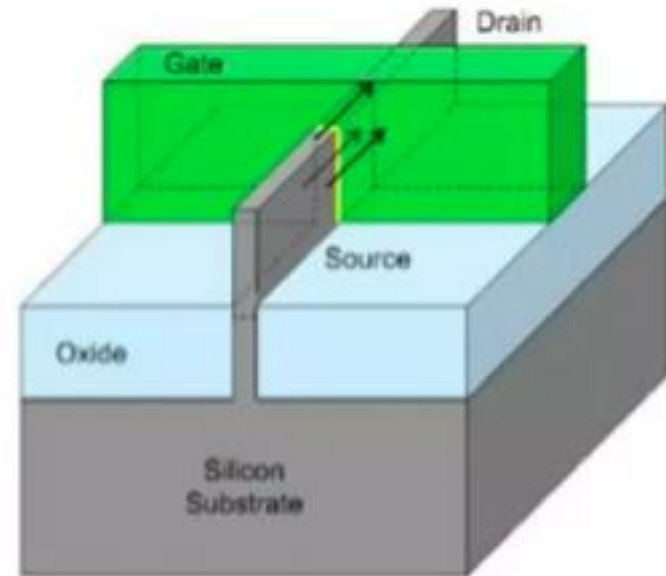
Lateral Planar vs. 3D FinFET

Traditional Planar



Traditional 2-D planar transistor form a conducting channel in the silicon region under the gate electrode when in the "on" state

3D FinFET



3-D Tri-Gate transistor form conducting channels on three sides of a vertical fin structure, providing "fully depleted" operation

CMOSFET Structure

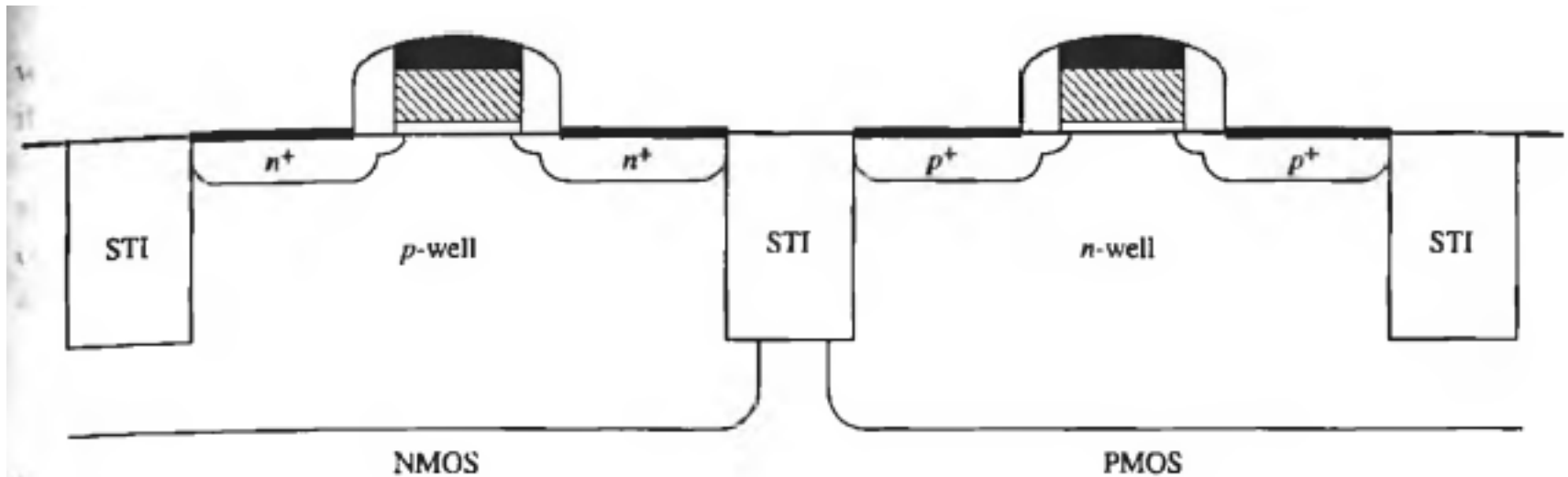
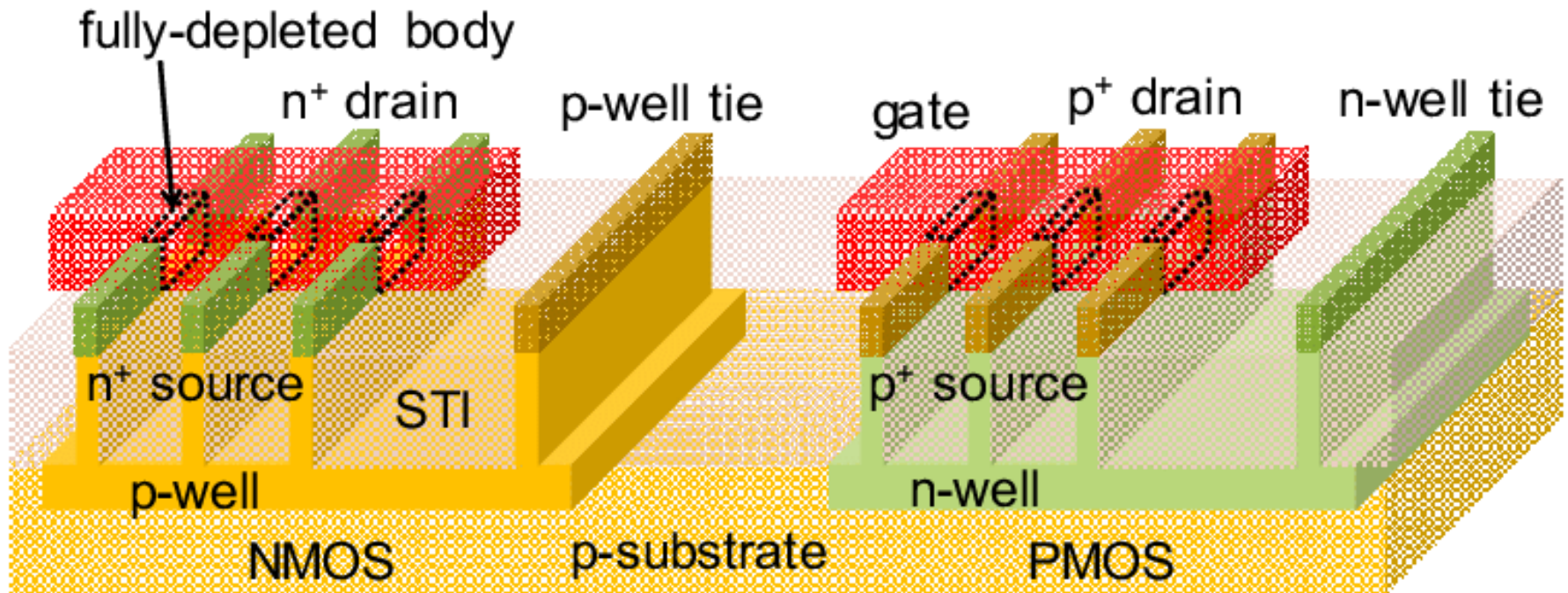


Figure 2.3
CMOS transistors structure.

CMOS FinFET Structure



Threshold Voltage

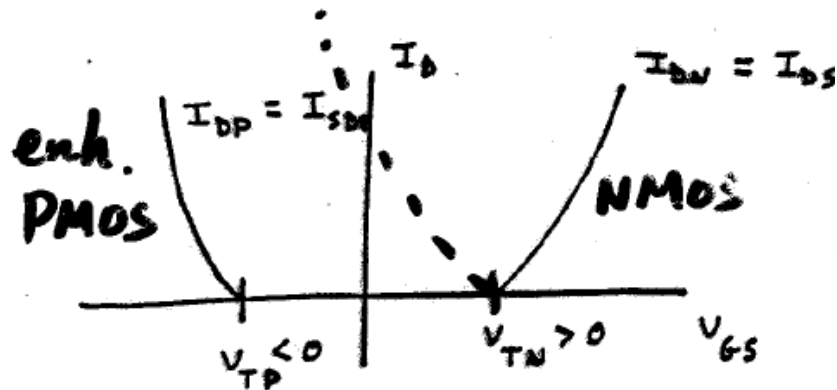


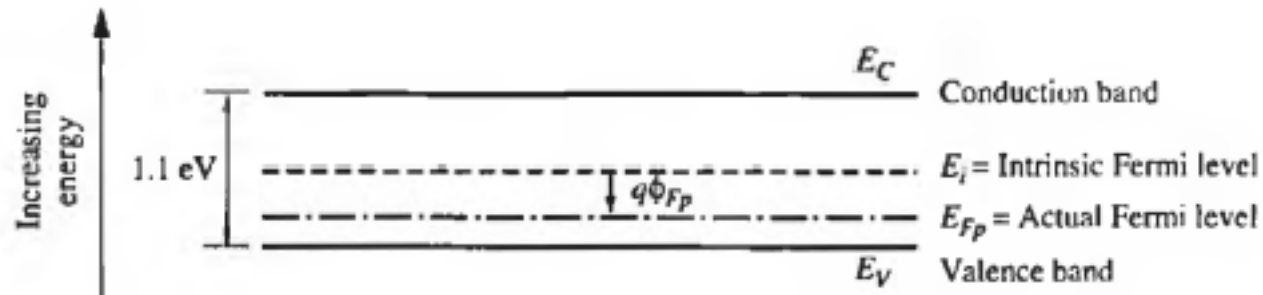
Table 2.1

Signs in threshold voltage equation

Parameter	NMOS	PMOS
Substrate	<i>p</i> -type	<i>n</i> -type
V_{T0}	+	-
ϕ_{GC} :		
n^+ polysilicon gate	-	-
p^+ polysilicon gate	+	+
Φ_F	-	+
Q_{B0}, Q_B	-	+
Q_{ox}	+	+
γ	+	-
X_d, C_{ox}	+	+
V_{SB}	+	-

Threshold Voltage

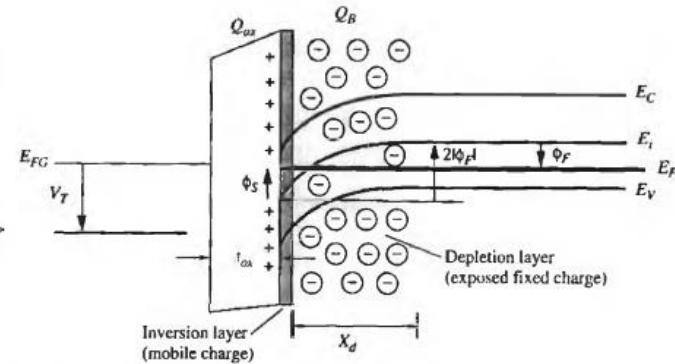
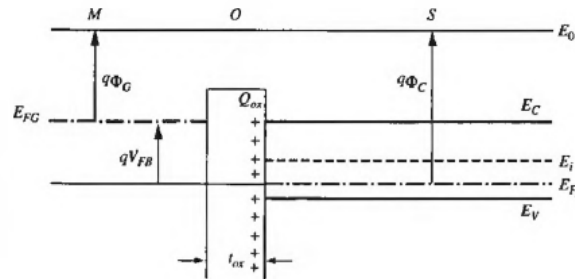
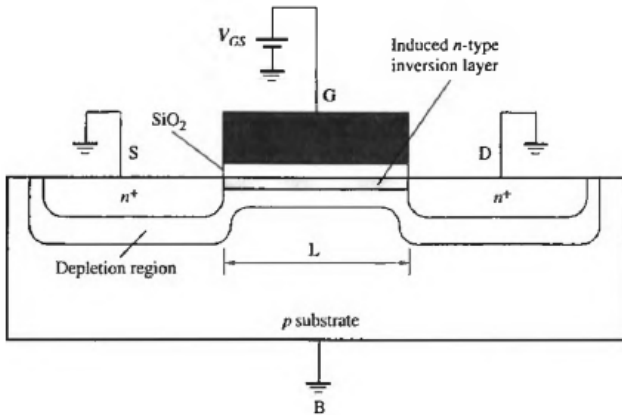
- Semiconductor Energy Band
 - Doping concentration $\longleftrightarrow$ Bulk Fermi Level



$$\phi_{Fp} = \frac{kT}{q} \ln \frac{n_i}{p} = \frac{kT}{q} \ln \frac{n_i}{N_A} < 0 \text{ (NMOS)}$$

$$\phi_{Fn} = \frac{kT}{q} \ln \frac{n}{n_i} \text{ where } n \approx N_D > 0 \text{ (PMOS)}$$

Threshold Voltage



$$V_{FB} = \phi_{GC} - \frac{Q_{ox}}{C_{ox}}$$

- NMOSFET: Positive gate charges are balanced by
 1. Negative ionized acceptors, N_A , in depletion region
 2. Mobile electrons in channel inversion layer

$$\phi_S = 2|\phi_{FP}| \text{ for } n_{\text{channel}} = p_{\text{bulk}} = N_A \text{ (Strong inversion)}$$

Threshold Voltage Calculation (HJS Ex.2.1-2.4)

130nm Technology NMOSFET: Calculate V_T at $V_{SB} = 0$.

$N_A = 3 \times 10^{17} \text{ cm}^{-3}$, $t_{ox} = 2.2 \text{ nm}$, $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$ at 300K

Poly-Si gate doping $N_D = 10^{20} \text{ cm}^{-3}$,

$Q_{ox} = +2 \times 10^{10} \text{ q cm}^{-2}$

$$V_{T0} = \phi_{GC} + 2\phi_f - \frac{Q_{f0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

1. $\phi_{GC} = \phi_{G(\text{gate})} - \phi_{Fp} = -0.55 - 0.44 = -0.99 \text{ V}$

2. $2|\phi_{Fp}| = \frac{2kT}{q} \left| \ln \frac{p}{n_i} \right| = 2(0.026) \left| \ln \frac{3 \times 10^{17}}{1.45 \times 10^{10}} \right| \cong 0.88 \text{ V}$

Threshold Voltage Calculation

(H&J Ex.2.1-2.4)

3.
$$Q_{B0} = -\sqrt{2qN_A\epsilon_{si}|-2\phi_F|}$$

$$Q_{B0} = -(2 \times 1.6 \times 10^{-19} \times (3)10^{17} \times 1.0 \times 10^{-12} \times |-0.88|)^{1/2}$$

$$\approx -3 \times 10^{-7} \text{ C/cm}^2$$

$$\epsilon_{ox} = 4\epsilon_0 = 3.5 \times 10^{-13} \text{ F/cm} \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{(4)(8.85 \times 10^{-14})}{22 \text{ \AA}} = 1.6 \times 10^{-6} \text{ F/cm}^2$$

$$\frac{Q_{B0}}{C_{ox}} = \frac{3 \times 10^{-7}}{1.6 \times 10^{-6}} = 0.188 \text{ V}$$

4.
$$\frac{Q_{ox}}{C_{ox}} = \frac{2 \times 10^{10} \times 1.6 \times 10^{-19}}{1.6 \times 10^{-6}} = 0.002 \text{ V}$$

$$V_{T0} = -0.99 - (-0.88) - (-0.188) - 0.002 = +0.08 \text{ V}$$

- Slightly enhancement mode ($V_T > 0$), needs a higher, more positive threshold voltage (0.4V)
- How to fix?

Threshold Voltage Calculation (H&J Ex.2.1-2.4)

Choice of N+ Poly-Si (vs. P+ Poly-Si):

$$\phi_{\text{FG(N+Poly-Si)}} = 0.55\text{V}; \phi_{\text{FG(P+Poly-Si)}} = -0.55\text{V}$$

Replacing N+ Poly-Si with P+ Poly-Si results in

$$\Delta\phi_{\text{FG}} = 1.10\text{V}$$

Hence, $V_{\text{TO}} = 1.10 + 0.08 = 1.18\text{V}$ **Too High!**

Targeted $V_{\text{TO}} = 0.4\text{V}$

Threshold Adjust Implant (H&J Ex.2-5)

Desired NFET $V'_{T0} = 0.4V$. What is N_I needed?

Implant acceptor ions to increase NFET V_{T0}

$$Q_I = q N_I = C_{ox} (V'_{T0} - V_{T0}) = C_{ox} \Delta V_{T0}$$

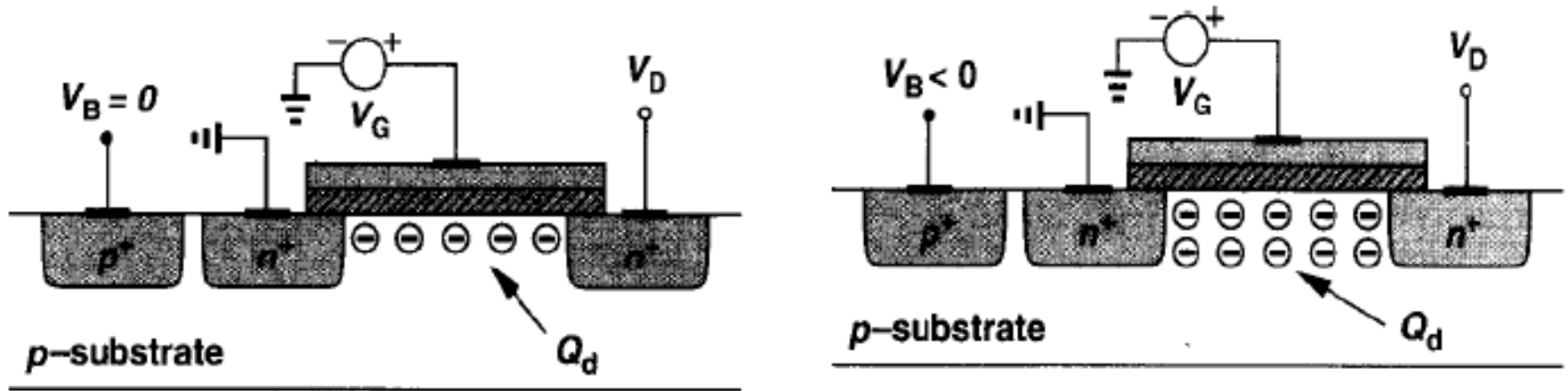
We know that V_T will be shifted by $Q_I / C_{ox} = qN_I / C_{ox}$

$$\therefore N_I = \frac{Q_I}{q} = \frac{C_{ox} \Delta V}{q} = \frac{C_{ox} (0.4 - (+0.08))}{1.6 \times 10^{-19}} = 3.2 \times 10^{12} \frac{\text{ions}}{\text{cm}^2}$$

Shallow ($0.1\mu\text{m}$ deep) box implant profile:

$$\begin{aligned} N_A = N_I / x_j &= (3.2 \times 10^{12} \text{cm}^{-2}) / (0.1 \times 10^{-4} \text{cm}) \\ &= 3.2 \times 10^{17} \text{cm}^{-3} \end{aligned}$$

Body Bias Effect



- Reverse biasing source/substrate junction increases space charges under channel in the substrate, hence increasing threshold voltage

$$V_{T0} + \gamma(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|})$$

where $\gamma = \frac{1}{C_{ox}} \sqrt{2q\epsilon_{si} N_A}$ body-effect coefficient

Body Bias Effect (HJS Ex. 2.3+)

- Since $\gamma = \sqrt{2q\epsilon_s N_A / C_{ox}}$

$$\gamma = \sqrt{(2)(1.6 \times 10^{-19})(11.7)(8.85 \times 10^{-14})(3 \times 10^{17}) / 1.6 \times 10^{-6}}$$

$$\approx 0.2 \text{ V}^{1/2}$$

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

$$= \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

$$= V_{T0} + \gamma(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|})$$

When $V_{SB} = V_{DD} = 1.2\text{V}$,

$$V_T = 0.08 + 0.2 [(1.2+0.88)^{1/2} - (0.88)^{1/2}] = 0.18\text{V}$$

Or $V_T = 0.5\text{V}$ with threshold adjust implant